US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250276338
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	LIU; Yang et al.

COATING DEVICE

Abstract

The present disclosure discloses a coating device. The coating device includes a coating roller, a scraping roller, and a blocking assembly. The scraping roller and the coating roller are arranged side by side, and a gap is provided between the scraping roller and the coating roller to allow a slurry to pass through. The blocking assembly is arranged at an outer side of the coating roller and an outer side of the scraping roller. The blocking assembly includes a support and a blocker mounted at the support. The blocker includes a cooperation portion cooperating with the coating roller and an engagement portion cooperating with the scraping roller. A first end of the cooperation portion and a first end of the engagement portion are connected and form an angle portion. The angle portion extends into the gap.

Inventors: **LIU; Yang (Xiamen, CN), WANG; Hao (Xiamen, CN)**

Applicant: **XIAMEN HITHIUM ENERGY STORAGE TECHNOLOGY CO., LTD.** (Xiamen, CN)

Family ID: **93912110**

Appl. No.: **19/062039**

Filed: **February 25, 2025**

Foreign Application Priority Data

CN	202420391646.6	Feb. 29, 2024

Publication Classification

Int. Cl.: **B05C1/08** (20060101)

U.S. Cl.:

CPC **B05C1/0865** (20130101); **B05C1/0882** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Chinese Patent Application No. 202420391646.6, filed on Feb. 29, 2024, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates to the field of coating technologies, and particularly, to a coating device.

BACKGROUND

[0003] The coating device is mainly used for production of a surface coating process of metallic foil, film, etc., generally adopts a special multifunctional coating device head, and can realize various forms of surface coating process. rolled substrates, such as metallic foil, film, cloth, paper, etc., are coated with a layer of ingredients with specific functions, and then wind it up after drying. Due to a high degree of automation, the coating device is widely used in coating production.

[0004] The most important and critical processes in the battery industry are slurry mixing, coating, and liquid injection, and the coating is the top priority, which is a technical focus of the entire industry. Quality of the coating directly affects safety and quality of a product. Currently, most battery coatings adopt coating devices, which include coating rollers and scraping rollers. During a coating process, a slurry is prone to spread outwards as the coating roller rotates, which affects a coating effect and also causes slurry waste.

SUMMARY

[0005] In view of the above problems, the present disclosure provides a coating device, capable of solving a problem in which a slurry is prone to spread outwards, which affects a coating effect and also causes slurry waste.

[0006] In a first aspect, the present disclosure provides a coating device. The device includes a coating roller, a scraping roller, and a blocking assembly. The scraping roller and the coating roller are arranged side by side, and a gap is provided between the scraping roller and the coating roller to allow a slurry to pass through. The blocking assembly is arranged at an outer side of the coating roller and an outer side of the scraping roller. The blocking assembly includes a support and a blocker mounted at the support. The blocker includes a cooperation portion cooperating with the coating roller and an engagement portion cooperating with the scraping roller. A first end of the cooperation portion and a first end of the engagement portion are connected with each other and form an angle portion, and the angle portion extends into the gap. A second end of the cooperation portion is located away from a second end of the engagement portion to form a first angle between the cooperation portion and the engagement portion, and the first angle is adjustable.

[0007] In the coating device of the above technical solution, by extending the angle portion of the blocker into the gap formed by the scraping roller and the coating roller, during a coating process, the slurry spreading outwards caused by the rotation of the coating roller can be blocked by the angle portion and fall into a slurry tank, which can ensure the coating effect on the one hand and avoid the slurry waste on the other hand. At the same time, since the first angle between the cooperation portion and the engagement portion is adjustable, the angle portion can extend into the gap to fill the gap and block the slurry from spreading outwards, no matter what reason for the change in the gap between the scraping roller and the coating roller. That is, the blocking assembly can constantly block the slurry from spreading outwards, and therefore the coating effect can be guaranteed when the gap changes.

[0008] As an optional technical solution of the present disclosure, the blocker further includes a first adjustment portion. Two opposite ends of the first adjustment portion being connected to the cooperation portion and the engagement portion, respectively, and the first adjustment portion is configured, by means of multi-stage extension and retraction, adjust the first angle according to a size of the gap.

[0009] In the above technical solution, the blocker adjusts the first angle through using the first adjustment portion, ensuring that when the size of the gap between the scraping roller and the coating roller changes, the angle portion may extend into the gap to fill the gap and prevent the slurry from spreading outwards.

[0010] As an optional technical solution of the present disclosure, the support includes a first support portion and a second support portion. The first support portion is connected to the engagement portion at a first end of the first support portion. The second support portion is connected to the cooperation portion at a first end of the second support portion. A second end of the first support portion and a second end of the second support portion are connected with each other and have a second angle. The second angle is adjustable to change the first angle.

[0011] In the above technical solution, the first support portion is connected to the engagement portion, and the second support portion is connected to the cooperation portion. The second angle between the first support portion and the second support portion that are connected to each other may be adjusted, enabling the first angle to be changed. Likewise, this arrangement can also ensure that when the size of the gap between the scraping roller and the coating roller changes, the angle portion may extend into the gap to fill the gap and prevent the slurry from spreading outwards.

[0012] As an optional technical solution of the present disclosure, the support further includes a second adjustment portion. Two opposite ends of the second adjustment portion are connected to the first support portion and the second support portion, respectively. The second adjustment portion is configured, by means of multi-stage extension and retraction, adjust the second angle according to the size of the gap, to change the first angle between the cooperation portion and the engagement portion.

[0013] In the above technical solution, the support adjusts the second angle through using the second adjustment portion, thereby achieving an adjustment of the first angle. Therefore, it is ensured that when the size of the gap between the scraping roller and the coating roller changes, the angle portion may still extend into the gap to fill the gap and prevent the slurry from spreading outwards.

[0014] As an optional technical solution of the present disclosure, the first adjustment portion or the second adjustment portion includes a housing, an elastic portion penetrating the housing, and at least two limit portions mounted at the housing in an extendable and retractable manner. The at least two limit portions are sequentially arranged in a length direction of the elastic portion, and the at least two limit portions are configured to selectively extend to a deformation path of the elastic portion relative to the housing or retract to be located outside the deformation path of the elastic portion relative to the housing, to match the first angle with the gap.

[0015] In the above technical solution, the first adjustment portion or the second adjustment portion adopts the multi-stage elastic retractable structure in such a manner that the first angle is adjusted through using the elastic portion, and the first angle is held to match the gap using the limit portion. In this way, a relatively simple structure can be used to adjust the first angle.

[0016] As an optional technical solution of the present disclosure, the first adjustment portion or the second adjustment portion includes a stator and a mover. The mover is extendable and retractable relative to the stator to adjust the first angle to match the gap.

[0017] In the above technical solution, the first adjustment portion or the second adjustment portion adopts the multi-stage rigid retractable structure in such a manner that the second angle is adjusted by using the extension and the retraction of the mover relative to the stator, thereby adjusting the first angle, such that the first angle to matches the size of the gap and hold the first angle to match the gap. In this way, a relatively simple structure can be used to adjust the first angle.

[0018] As an optional technical solution of the present disclosure, the cooperation portion has a first surface and a second surface opposite to the first surface of the cooperation portion, the first surface of the cooperation portion being an arc surface in contact with and in cooperation with the coating roller; and the engagement portion has a first surface and a second surface opposite to the first surface of the engagement portion, the first surface of the engagement portion being an arc surface in contact with and in cooperation with the scraping roller.

[0019] In the above technical solution, since the first surface of the cooperation portion is the arc surface; the first surface of the engagement portion is the arc surface; a peripheral surface of the coating roller is an arc surface; and a peripheral surface of the scraping roller is an arc surface, the cooperation portion and the coating roller can be seamlessly fitted, and the engagement portion and the scraping roller can also be seamlessly fitted. In this way, the slurry is avoided from entering a position between the cooperation portion and the coating roller or a position between the engagement portion and the scraping roller, which would otherwise result in the waste and affect the coating effect.

[0020] As an optional technical solution of the present disclosure, the blocker further includes a connection portion extending from the second surface of the cooperation portion; the connection portion and the engagement portion are located at a side of the cooperation portion where the second surface of the cooperation portion is located; the connection portion is configured to connect to an external structure; the cooperation portion, the engagement portion, and the connection portion together form an engagement groove; and the support is disposed in the engagement groove.

[0021] In the above technical solution, the arrangement of the connection portion can facilitate a fixed mounting of the entire blocker, and the engagement groove formed by the cooperation portion, the engagement portion, and the connection portion can provide a satisfactory mounting of the support.

[0022] As an optional technical solution of the present disclosure, the coating device further includes a loader connected to the connection portion.

[0023] In the above technical solution, the mounting of the blocking assembly is achieved by the loader connected to the connection portion. In this way, the mounting is simple and convenient.

[0024] As an optional technical solution of the present disclosure, the coating device further includes a clamping plate configured to fixedly clamp the support and the connection portion to the loader.

[0025] In the above technical solution, the arrangement of the clamping plate can firmly and fixedly clamp the support and the loader without a movement. Therefore, the angle portion of the blocker can cooperate well with the coating roller and the scraping roller. Thus, the effect of blocking the overflowing of the slurry is improved.

[0026] As an optional technical solution of the present disclosure, the blocker is an ethylene-vinyl acetate copolymer, EVA, foam board, an acrylic board, a plastic board, or a metallic board.

[0027] In the above technical solution, when the blocker is the EVA foam board, the angle portion of the blocker may extend into the gap to be in good cooperation with the scraping roller and the coating roller, which can not only block the overflowing of the slurry but also provide a sound insulation effect. Additionally, the slurry can be prevented from leaking and seeping into a space between the foam and the coating roller, which causes the formation of particle agglomerations and dark marks on the film surface. That is, the issue of dark marks caused by the slurry seepage can also be improved.

[0028] As an optional technical solution of the present disclosure, the coating device further includes: a first support frame and a first connector, the scraping roller being rotatably connected to the first support frame through the first connector; and a second support frame and a second connector, the coating roller being rotatably connected to the second support frame through the second connector.

[0029] In the above technical solution, the first support frame and the first connector jointly realize the mounting and the fixing of the scraping roller, and the second support frame and the second

connector jointly realize the mounting and the fixing of the coating roller. As a result, the scraping roller and the coating roller can be arranged side by side.
[0030] The above description is only an overview of the technical solutions of the present disclosure. In order to have a clearer understanding of the technical means of the present disclosure, the technical solutions can be implemented in accordance with the contents of the specification. Moreover, in order to make the above and other purposes, features and advantages of the present disclosure more obvious and easier to understand, specific embodiments of the present disclosure are given below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] Various other advantages and benefits will become apparent to those skilled in the art after reading the detailed description of preferred embodiments given below. The accompanying drawings are used for a purpose of illustrating the preferred embodiments only, rather than limiting the present disclosure. Moreover, throughout the accompanying drawings, same elements are denoted by same reference numerals.
[0032] FIG. **1** is a schematic structural plan view of a coating device according to some embodiment of the present disclosure.
[0033] FIG. **2** is another schematic structural plan view of a coating device according to some embodiments of the present disclosure.
[0034] FIG. **3** is yet another schematic structural plan view of a coating device according to some embodiments of the present disclosure.
[0035] FIG. **4** is a schematic structural view of a blocker of a blocking assembly in a coating device according to some embodiments of the present disclosure.
[0036] FIG. **5** is a schematic structural view of a blocker and a first adjustment portion of a coating device according to some embodiments of the present disclosure.
[0037] FIG. **6** is a schematic structural view of a blocker and another first adjustment portion of a coating device according to some embodiments of the present disclosure.
[0038] FIG. **7** is a schematic structural view of a blocker and a second adjustment portion of a coating device according to some embodiments of the present disclosure.
[0039] FIG. **8** is a schematic structural view of a blocker and another second adjustment portion of a coating device according to some embodiments of the present disclosure.
[0040] Reference numerals in specific embodiments are as follows: [0041] coating device **100**; [0042] coating roller **10**, scraping roller **30**, gap **40**, center line M, blocking assembly **50**; [0043] support **51**, first support portion **511**, first end **5111** of first support portion, second end **5113** of first support portion, second support portion **513**, first end **5131** of second support portion, second end **5133** of second support portion, second adjustment portion **515**, second housing **5151**, second elastic portion **5153**, second limit portion **5155**, second stator **5157**, second mover **5159**; [0044] blocker **53**, cooperation portion **531**, first end **5311** of cooperation portion, second end **5313** of cooperation portion, first surface **5315** of cooperation portion, second surface **5317** of cooperation portion, engagement portion **533**, first end **5331** of engagement portion, the second end **5333** of engagement portion, first surface **5335** of engagement portion, second surface **5337** of engagement portion, angle portion **535**, first adjustment portion **537**, first housing **5371**, first elastic portion **5373**, first limit portion **5375**, first stator **5377**, first mover **5379**, first angle α, second angle β, connection portion **539**, engagement groove **530**; [0045] loader **55**; [0046] clamping plate **57**, first clamping arm **571**, second clamping arm **573**, connection arm **575**; [0047] first support frame **61**, first connector **71**, second support frame **63**, second connector **73**.
DETAILED DESCRIPTION

[0048] Embodiments of technical solutions of the present disclosure will be described in detail below with reference to the accompanying drawings. The following embodiments are only used to more clearly illustrate the technical solutions of the present disclosure, and therefore are only used as examples, and cannot be used to limit the protection scope of the present disclosure.

[0049] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art of the present disclosure. Terms in the specification of the present disclosure herein are only used for the purpose of describing exemplary embodiments, and are not intended to limit the present disclosure. Terms "including" and "having" and any variants thereof as used in the description of the embodiments of the present disclosure, the appended claims, and the above accompanying drawings are intended to cover non-exclusive inclusions.

[0050] In the description of the present disclosure, terms such as "first" and "second" are used herein for purposes of description and are not intended to indicate or imply relative importance, or to implicitly show the number of technical features, specific orders, or primary and secondary relationships indicated. In the description of the present disclosure, "a plurality of" means two or more, unless otherwise specifically defined.

[0051] Reference herein to "an embodiment" means that a particular feature, structure or characteristic described in combination with the embodiment can be included in at least one embodiment of the present disclosure. The appearance of this phrase in various places in the specification are not necessarily all referring to the same embodiment, nor is it an independent or alternative embodiment that is mutually exclusive of other embodiments. Those skilled in the art understand, both explicitly and implicitly, that the embodiments described herein may be combined with other embodiments.

[0052] In the description of the present disclosure, "and/or" describes an association relationship between correlated objects, including three relationships. For example, "A and/or B" can mean A only, B only, or both A and B. In addition, the symbol "/" generally indicates an "or" relationship between the correlated objects preceding and succeeding the symbol.

[0053] In the description of the present disclosure, the term "a plurality of" refers to more than two (including two). Similarly, "a plurality of groups" refers to more than two groups (including two groups), and "a plurality of pieces" refers to more than two pieces (including two pieces).

[0054] In the description of the present disclosure, it is to be understood that, terms such as "center," "longitudinal," "lateral," "length," "width," "thickness," "over," "below," "front," "back," "left," "right," "vertical," "horizontal," "top," "bottom," "in," "out," "clockwise," "counterclockwise," "axial," "radial," "circumferential," etc., is based on the orientation or position relationship shown in the accompany drawings, and is only for the convenience of describing the present disclosure and simplifying the description, rather than indicating or implying that the associated device or element must have a specific orientation, or be constructed and operated in a specific orientation, and therefore cannot be understood as a limitation on the present disclosure.

[0055] In the description of the present disclosure, unless specified or limited otherwise, the terms "mounted," "connected," "coupled" and "fixed" are understood broadly, such as fixed, detachable mountings, connections and couplings or integrated, and can be mechanical or electrical mountings, connections and couplings, and also can be direct and via media indirect mountings, connections, and couplings, and further can be inner mountings, connections and couplings of two components or interaction relations between two components.

[0056] Referring to FIGS. **1** and **2**, the present disclosure provides a coating device **100**. The coating device **100** includes a coating roller **10**, a scraping roller **30**, and a blocking assembly **50**. The scraping roller **30** and the coating roller **10** are arranged side by side, and a gap **40** is provided between the scraping roller **30** and the coating roller **10** to allow a slurry to pass through. The blocking assembly **50** is arranged at an outer side of the coating roller **10** and an outer side of the scraping roller **30**. The blocking assembly **50** includes a support **51** and a blocker **53** mounted at the support **51**. Referring to FIG. **4**, the blocker **53** includes a cooperation portion **531** cooperating with the coating roller **10** and an engagement portion **533** cooperating with the scraping roller **30**. A first end **5311** of the cooperation portion **531** and a first end **5331** of the engagement portion **533** are connected with each other and form an angle portion **535**. A second end **5313** of the cooperation portion **531** is located away from a second end **5333** of the engagement portion **533** with each other to form a first angle α between the cooperation portion **531** and the engagement portion **533**. The angle portion **535** extends into the gap **40**. The first angle α between the cooperation portion **531** and the engagement portion **533** is adjustable.

[0057] The coating device **100** is a device for coating rolled substrates, such as metallic foil, film, cloth, paper, etc., with a layer of ingredients with specific functions, and then drying and rewinding. In some embodiments of the present disclosure, the coating device **100** is a device for coating a rolled metallic foil with a slurry to form a coating region of an electrode sheet. It should be noted that in other embodiments, the coating device **100** may also be a device for coating other ingredients with specific functions onto a substrate to form other elements.

[0058] The coating roller **10** is a rotatable roller for winding a substrate (such as metallic foil). The coating roller **10** has two end surfaces and a circumferential surface connected to the two end surfaces. Typically, the coating roller **10** is in a cylindrical structure with two circular end surfaces and a peripheral surface that is a circumferential surface. The scraping roller **30** is a roller for extruding a slurry when the slurry is applied onto a substrate to achieve a required thickness of the slurry. The scraping roller **30** also has two end surfaces and a peripheral surface connected to each of the two end surfaces. In some embodiments, the scraping roller **30** is in a cylindrical structure with two circular end surfaces and a peripheral surface that is a circumferential surface, as shown in FIG. **2** or FIG. **3**. In some other embodiments, the scraping roller **30** is not in a cylindrical structure with two non-circular end surfaces, but a comma roller having a “comma” portion. That is, the end surface of the scraping roller **30** is in a “comma” shape. In this case, an extrusion force of the scraping roller **30** is more concentrated on the “comma” portion, resulting in more precise coating and a better effect.

[0059] The scraping roller **30** and the coating roller **10** are arranged side by side. In one embodiment, the scraping roller **30** and the coating roller **10** are arranged side by side in a direction of center line M of end surfaces at a side of the scraping roller **30** and the coating roller **10**. For example, the scraping roller **30** and the coating roller **10** are arranged side by side up and down as shown in FIG. **2** or FIG. **3**. A gap **40** is provided between the scraping roller **30** and the coating roller **10** to allow the slurry to pass through, that is, the gap **40** is formed by the peripheral surface of the scraping roller **30** and the peripheral surface of the coating roller **10** and is typically the required thickness of the formed coating region. When the slurry is coated through the gap **40** between the scraping roller **30** and the coating roller **10**, the scraping roller **30** scrapes off an excess slurry, and the excess slurry scraped off flows back into the slurry tank.

[0060] The blocking assembly **50** is an assembly for preventing the slurry from overflowing. In the present disclosure, two slurry blocking assemblies **50** are provided. The two slurry blocking assemblies **50** are arranged at two ends of a substrate in a width direction of the substrate and configured to prevent the slurry from overflowing from an edge of the substrate in the width direction of the substrate. The blocking assembly **50** includes a support **51** and a blocker **53** mounted at the support **51**. The blocker **53** is specifically configured to prevent the slurry from overflowing, and the support **51** is configured to support the blocker **53**.

[0061] In some embodiments, the blocker **53** is an EVA foam board, an acrylic board, a plastic board, or a metallic board. The EVA foam board has good softness and elasticity, may still have good flexibility at a temperature of −50° C., good chemical stability, good aging resistance and ozone resistance, and is harmless. Therefore, when the blocker **53** is the EVA foam board, the angle portion **535** of the blocker **53** may extend into the gap **40** to be in good cooperation with the scraping roller **30** and the coating roller **10**, which can not only block the overflowing of the slurry but also provide a sound insulation effect. Additionally, the slurry can be prevented from leaking out and seeping into a space between the blocker **53** (the EVA foam board) and the coating roller **10**, which causes the formation of particle agglomerations and dark marks on the film surface. That is, the issue of dark marks caused by the slurry seepage can also be improved.

[0062] In addition, the EVA foam board may be formed through stamping, injection molding, extrusion molding, blow molding, calendering, rotational vacuum thermoforming, foaming, coating, heat sealing, welding, and other molding processes. Using a standard EVA foam board after stamping can ensure neat trimming of the EVA foam board, thereby ensuring that the neat trimming of the EVA foam board is consistent with a side edge of the coating layer. Moreover, a tip part of the angle portion **535** is adjusted from a previous rounded corner to a suitable sharp corner, which can avoid an uneven grinding surface caused by the scraping roller **30** grinding the EVA foam board and causing the dark marks.

[0063] Referring to FIGS. **2** and **4**, or FIGS. **3** and **4**, the blocker **53** includes a cooperation portion **531** and an engagement portion **533** connected to the cooperation portion **531** of the blocker **53**. The cooperation portion **531** cooperates with the coating roller **10**, and the engagement portion **533** cooperates with the scraping roller **30**. The cooperation portion **531** is in a curved structure and has a first end **5311** and a second end **5313** that are opposite to each other. The cooperation portion **531** is also a curved structure and also has a first end **5331** and a second end **5333** that are opposite to each other. A first end **5311** of the cooperation portion **531** is connected to a first end **5331** of the engagement portion **533**, and a second end **5313** of the cooperation portion **531** is located away from a second end **5333** of the engagement portion **533**, such that the cooperation portion **531** and the engagement portion **533** together form a “chevron” structure that has an angle portion **535** with a triangular or trapezoidal cross section at a connection end of the cooperation portion **531** and the engagement portion **533**. A first angle α between the cooperation portion **531** and the engagement portion **533**. The first angle α is an angle between two intersecting surfaces of the cooperation portion **531** and the engagement portion. When the two intersecting surfaces are arc surfaces, the first angle α is an angle between tangents of the two arc surfaces passing through an intersection point (an intersection point of the arc surfaces), as shown in FIG. **2**. The angle portion **535** extends into the gap **40**, and the first angle α between the cooperation portion **531** and the engagement portion **533** is adjustable, that is, the first angle α is variable, for example, from 70° to 65°, 60°, 55°, 50°, 45°, 40°, or 30°, etc., which will not be enumerated herein.

[0064] In one embodiment, the cooperation portion **531** includes a first surface **5315** and a second surface **5317** opposite to the first surface **5315**, and the first surface **5315** of the cooperation portion **531** is an arc surface in contact with and in cooperation with the coating roller **10**. The engagement portion **533** also includes a first surface **5335** and a second surface **5337** opposite to the first surface **5335**, and the first surface **5335** of the engagement portion **533** is an arc surface in contact with and in cooperation with the scraping roller **30**. Since the first surface **5315** of the cooperation portion **531** is an arc surface, and the peripheral surface of the coating roller **10** is a circumferential surface that is also an arc surface, the cooperation portion **531** and the coating roller **10** may fit well together. In this way, during the coating process, rotation of the coating roller **10** is not hindered by the cooperation portion **531**. At the same time, the cooperation portion **531** can better wrap the peripheral surface of the coating roller **10** to achieve seamless fitting, thereby avoiding the slurry from entering a position between the cooperation portion **531** and the coating roller **10**, which would otherwise result in the waste and affect the coating effect. Similarly, since the first surface **5335** of the engagement portion **533** is an arc surface, and the peripheral surface of the scraping roller **30** is a circular surface that is also an arc surface, the engagement portion **533** and the scraping roller **30** may fit well together. In this way, during the coating process, rotation of the scraping roller **30** is not hindered by the engagement portion **533**. At the same time, the engagement portion **533** can better wrap the circumferential surface of the scraping roller **30** to achieve seamless fitting, thereby avoiding the slurry from entering a position between the engagement portion **533** and the scraping roller **30**, which would otherwise result in the waste and affect the coating effect.

[0065] In the coating device **100** of the above-mentioned technical solution, by extending the angle portion **535** of the blocker **53** into the gap **40** formed by the scraping roller **30** and the coating roller **10**, during the coating process, the slurry spreading outwards caused by the rotation of the coating roller **10** can be blocked by the angle portion **535** and fall into the slurry tank, which can ensure the coating effect on the one hand and avoid the slurry waste on the other hand. At the same time, since the first angle α between the cooperation portion **531** and the engagement portion **533** is adjustable, the angle portion **535** can extend into the gap **40** to fill the gap **40** and block the slurry from spreading outwards, no matter what reason for the change in the size of the gap **40** between the scraping roller **30** and the coating roller **10**. That is, the blocking assembly **50** can constantly block the slurry from spreading outwards, and therefore the coating effect can be guaranteed when the gap **40** changes.

[0066] Further, referring to FIGS. **2** and **4**, the blocker **53** may further include a first adjustment portion **537** with a multi-stage retractable structure. Two opposite ends of the first adjustment portion **537** are connected to the cooperation portion **531** and the engagement portion **533**, respectively. The first adjustment portion **537** is configured, by means of multi-stage extension and retraction, adjust the first angle α between the cooperation portion **531** and the engagement portion **533** according to the size of the gap **40**, such that the first angle α matches the gap **40**.

[0067] Referring to FIG. **5**, in some embodiments, the first adjustment portion **537** may be of a multi-stage retractable structure that is elastically deformable (hereinafter referred to as a multi-stage elastic retractable structure), and the first adjustment portion **537** has an end connected to the cooperation portion **531** and another end connected to the engagement portion **533**. In one embodiment, the first adjustment portion **537** (the multi-stage elastic retractable structure) includes a first housing **5371**, a first elastic portion **5373** penetrating the first housing **5371**, and at least two first limit portions **5375** mounted at the first housing **5371** in an extendable and retractable manner. Two opposite ends of the first elastic portion **5373** are connected to the cooperation portion **531** and the engagement portion **533**, respectively, to apply a stretching force or a tensioning force to the cooperation portion **531** and/or the engagement portion **533**, thereby stretching or tensioning the cooperation portion **531** and/or the engagement portion **533**. The at least two first limit portions **5375** are sequentially arranged in a length direction of the first elastic portion **5373** and configured to selectively extend to a deformation path of the first elastic portion **5373** relative to the first housing **5371** or retract to be located outside the deformation path of the first elastic portion **5373** relative to the first housing **5371**, such that the first angle matches the size of the gap **40** when a magnitude of the stretching force or a magnitude of the tensioning force of the first elastic portion **5373** changes. That is, the angle portion **535** can constantly be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. That is, the first surface **5315** of the cooperation portion **531** constantly maintains a good fit with the peripheral surface of the coating roller **10**, and the first surface **5335** of the engagement portion **533** constantly maintains a good fit with the peripheral surface of the scraping roller **30**. When the external force applied to the cooperation portion **531** and/or the engagement portion **533** changes, the change in the external force is transmitted to the first adjustment portion **537**. In this case, the stretching force or the tensioning force of the first elastic portion **5373** of the first adjustment portion **537** changes, and the first limit portion **5375** initially cooperating with the first elastic portion **5373** retracts due to the external force and is located outside the deformation path of the first elastic portion **5373**. The first elastic portion **5373** deforms to adjust the first angle α to enable that the first angle α matches the size of the gap **40**. Another first limit portion **5375** of the first adjustment portion **537** may automatically extend to the deformation path of the first elastic portion **5373** relative to the first housing **5371** to hold the cooperation portion **531** and the engagement portion **533** at a position where the first angle α matches the size of the gap **40**. For example, when the gap **40** between the scraping roller **30** and the coating roller **10** increases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** decreases. In this case, the stretching force of the first adjustment portion **537** increases or the tensioning force of the first adjustment portion **537** decreases, and the first limit portion **5375** initially cooperating with the first elastic portion **5373** retracts due to the external force and is located outside the deformation path of the first elastic portion **5373**. The first elastic portion **5373** of the multi-stage elastic retractable structure extends, and the first angle α between the cooperation portion **531** and the engagement portion **533** increases and rematches the increased gap **40**. In this way, the angle portion **535** can still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. For another example, when the gap **40** between the scraping roller **30** and the coating roller **10** decreases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** increases. In this case, the stretching force of the first adjustment portion **537** decreases or the tensioning force of the first adjustment portion **537** increases, and the first limit portion **5375** initially cooperating with the first elastic portion **5373** retracts due to the external force and is located outside the deformation path of the first elastic portion **5373**. The first elastic portion **5373** of the multi-stage elastic retractable structure retracts, and the first angle α between the cooperation portion **531** and the engagement portion **533** decreases and rematches the decreased gap **40**. In this way, the angle portion **535** may still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing.

[0068] Referring to FIG. **6**, in other embodiments, the first adjustment portion **537** may be of a multi-stage retractable structure without elastic deformation (hereinafter referred to as a multi-stage rigid retractable structure), such as a multi-stage cylinder retractable structure, a multi-stage hydraulic cylinder retractable structure, or a multi-stage linear retractable motor. The first adjustment portion **537** has an end connected to the cooperation portion **531** and another end connected to the engagement portion **533**. The first adjustment portion **537** (the multi-stage rigid retractable structure) includes a first stator **5377** and a first mover **5379**. One of the first stator **5377** and the first mover **5379** is connected to the cooperation portion **531**, and another one of the first stator **5377** and the first mover **5379** is connected to the engagement portion **533**. The first mover **5379** is retractable relative to the first stator **5377** to stretch the cooperation portion **531** and the engagement portion **533**. When the external force applied on the cooperation portion **531** and/or the engagement portion **533** changes, the first adjustment portion **537** can automatically adjust a degree of the stretching between the cooperation portion **531** and the engagement portion **533** in response to the change of the external force, to adjust the first angle α between the cooperation portion **531** and the engagement portion **533** and hold the cooperation portion **531** and the engagement portion **533** at a position where the first angle constantly matches the size of the gap **40**. For example, when the gap **40** between the scraping roller **30** and the coating roller **10** increases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** decreases. In this case, the first mover **5379** extends relative to the first stator **5377** to increase the first angle α and rematch the increased gap **40**. In this way, the angle portion **535** may still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. For another example, when the gap **40** between the scraping roller **30** and the coating roller **10** decreases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** increases. In this case, the first mover **5379** retracts relative to the first stator **5377** to allow the first angle α to decrease and rematch the reduced gap **40**. In this way, the angle portion **535** may still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing.

[0069] In the above technical solution, the blocker **53** uses the first adjustment portion **537** to adjust the first angle α, which is simple in structure and easy to implement, and can ensure that when the size of the gap **40** between the scraping roller **30** and the coating roller **10** changes, the angle portion **535** can still extend into the gap **40** to fill the gap **40** and prevent the slurry from spreading outwards. Further, when the first adjustment portion **537** adopts the first the multi-stage elastic retractable structure, the first angle is adjusted by using the elastic portion, and the first angle is held to match the gap using the limit portion to **40**. In this way, a relatively simple structure can be used to adjust the first angle α. When the first adjustment portion **537** adopts the multi-stage rigid retractable structure, the first angle α may be adjusted by using the extension and the retraction of the mover **5379** relative to the stator **5377**, and the first angle α is held to be matched with the gap **40**. In this way, a relatively simple structure can be used to adjust the first angle α.

[0070] Referring to FIGS. **2** and **4**, or FIGS. **3** and **4**, in some embodiments, the support **51** has an open structure with one end closed and another end opened, formed by two plate-like structures connected at one end and away from each other at another end. In one embodiment, the support **51** (open structure) may include a first support portion **511** and a second support portion **513**. In some examples, the first support portion **511** and the second support portion **513** are both plate-shaped. In other examples, the first support portion **511** and the second support portion **513** both have an elongated shape. In other embodiments, the first support portion **511** and the second support portion **513** may also have other shapes, which will not be enumerated herein. The first support portion **511** is connected to the engagement portion **533** at a first end **5111** of the first support portion **511**, and the second support portion **513** is connected to the cooperation portion **531** at a first end **5131** of the second support portion **513**. A second end **5113** of the first support portion **511** and a second end **5113** of the second support portion **513** are connected to form a second angle β, and the second angle β is adjustable to change the first angle α.

[0071] In the above technical solution, the first support portion **511** is connected to the engagement portion **533**, and the second support portion **513** is connected to the cooperation portion **531**. The first angle α may be changed by adjusting second angle β between the first support portion **511** and the second support portion **513** connected to the first support portion **511**, which can also ensure that when the size of the gap **40** between the scraping roller **30** and the coating roller **10** changes, the angle portion **535** may extend into the gap **40** to fill the gap **40** and prevent the slurry from spreading outwards.

[0072] Further, referring to FIGS. **3** and **4**, in some embodiments, the support **51** further includes a second adjustment portion **515** with a multi-stage retractable structure. Two opposite ends of the second adjustment portion **515** are connected to the first support portion **511** and the second support portion **513**, respectively. The second adjustment portion **515** is configured, by means of multi-stage extension and retraction, adjust the second angle β according to a size of the gap **40**, thereby enabling the first angle α to be changed, such that the first angle α matches the gap **40**.

[0073] Referring to FIG. **7**, in some embodiments, the second adjustment portion **515** may have a multi-stage retractable structure that is elastically deformable (hereinafter referred to as a multi-stage elastic retractable structure), the second adjustment portion **515** has an end connected to the first support portion **511** and another end connected to the second support portion **513**. In one embodiment, the second adjustment portion **515** (the multi-stage elastic retractable structure) includes a second housing **5151**, a second elastic portion **5153** penetrating the second housing **5151**, and at least two second limit portions **5155** mounted at the second housing **5151** in an extendable and retractable manner. Two opposite ends of the second elastic portion **5153** are connected to the first support portion **511** and the second support portion **513**, respectively, to apply a stretching force or a tensioning force to the first support portion **511** and/or the second support portion **513**, thereby stretching or tensioning the first support portion **511** and/or the second support portion **513**. The at least two second limit portions **5155** are sequentially arranged in a length direction of the second elastic portion **5153** and configured to selectively extend to a deformation path of the second elastic portion **5153** relative to the second housing **5151** or retract to be located outside the deformation path of the second elastic portion **5153** relative to the second housing **5151**, such that when a magnitude of the stretching force or a magnitude of the tensioning force of the second elastic portion **5153** changes, the second angle is adjusted, which in turn adjusts the first angle until the first angle matches the size of the gap **40**. That is, the angle portion **535** can constantly be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. That is, the second surface **5315** of the cooperation portion **531** constantly maintains a good fit with the peripheral surface of the coating roller **10**, and the second surface **5335** of the engagement portion **533** constantly maintains a good fit with the peripheral surface of the scraping roller **30**. When the external force applied to the cooperation portion **531** and/or the engagement portion **533** changes, the change in the external force is transmitted to the second adjustment portion **515** through the first support portion **511** and the second support portion **513**. In this case, the stretching force or tensioning force of the second elastic portion **5153** of the second adjustment portion **515** changes, and the second limit portion **5155** initially cooperating with the second elastic portion **5153** retracts due to the external force and is located outside the deformation path of the second elastic portion **5153**. The second elastic portion **5153** deforms to adjust the second angle β, which in turn adjusts the first angle α such that the first angle α matches the size of the gap **40**. At the same time, another second limit portion **5155** of the second adjustment portion **515** may automatically extend to the deformation path of the second elastic portion **5153** relative to the second housing **5151** to hold the first support portion **511** and the second support portion **513** at a position where the first angle α matches the size of the gap **40**. For example, when the gap **40** between the scraping roller **30** and the coating roller **10** increases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** decreases. In this case, the stretching force of the second adjustment portion **515** increases or the tensioning force of the second adjustment portion **515** decreases, and the second limit portion **5155** initially cooperating with the second elastic portion **5153** retracts due to the external force and is located outside the deformation path of the second elastic portion **5153**. The second elastic portion **5153** of the multi-stage elastic retractable structure extends, and the second angle β of the support **51** increases. Therefore, the first angle α between the cooperation portion **531** and the engagement portion **533** increases and rematches the increased gap **40**. In this way, the angle portion **535** can still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. For another example, when the gap **40** between the scraping roller **30** and the coating roller **10** decreases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** increases. In this case, the stretching force of the second adjustment portion **515** decreases or the tensioning force of the second adjustment portion **515** increases, and the second limit portion **5155** initially cooperating with the second elastic portion **5153** retracts due to the external force and is located outside the deformation path of the second elastic portion **5153**. The second elastic portion **5153** of the multi-stage elastic retractable structure retracts, and the second angle β of the support **51** decreases. Therefore, the first angle α also decreases and rematches with the decreased gap **40**. In this way, the angle portion **535** can still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing.

[0074] Referring to FIG. **8**, in some other embodiments, the second adjustment portion **515** may be of a multi-stage retractable structure without elastic deformation (hereinafter referred to as a multi-stage rigid retractable structure), such as a multi-stage cylinder retractable structure, a multi-stage hydraulic cylinder retractable structure, or a multi-stage linear retractable motor. The second adjustment portion **515** has an end connected to the first support portion **511** and another end connected to the second support portion **513**. The second adjustment portion **515** (the multi-stage rigid retractable structure) includes a second stator **5157** and a second mover **5159**. One of the second stator **5157** and the second mover **5159** is connected to the first support portion **511**, and another one of the second stator **5157** and the second mover **5159** is connected to the second support portion **513**. The second mover **5159** is retractable relative to the second stator **5157** to stretch the support **51**. When the external force applied to the cooperation portion **531** and/or the engagement portion **533** changes, the second adjustment portion **515** can automatically adjust a degree of the stretching between the second adjustment portion **515** and the support **51** in response to the change in the external force, to adjust the second angle β of the support **51** and hold the first support portion **511** and the second support portion **513** at a position where the first angle constantly matches the size of the gap **40**. For example, when the gap **40** between the scraping roller **30** and the coating roller **10** increases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** decreases, and the second mover **5159** extends relative to the second stator **5157** to increase the second angle β of the support **51**. Therefore, the first angle α between the cooperation portion **531** and the engagement portion **533** also increases and rematches the increased gap **40**, and the angle portion **535** can still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing. For another example, when the gap **40** between the scraping roller **30** and the coating roller **10** decreases, the external force applied to the cooperation portion **531** and/or the engagement portion **533** increases. In this case, the second mover **5159** retracts relative to the second stator **5157** to reduce the second angle β of the support **51**. Therefore, the first angle α between the cooperation portion **531** and the engagement portion **533** also decreases and rematches the decreased gap **40**. In this way, the angle portion **535** can still be filled into the gap **40** between the scraping roller **30** and the coating roller **10** to prevent the slurry from overflowing.

[0075] In the above technical solution, the support **51** adjusts the second angle β through using the second adjustment portion **515**, thereby adjusting the first angle α between the cooperation portion **531** and the engagement portion **533**. Therefore, it is ensured that when the size of the gap **40** between the scraping roller **30** and the coating roller **10** changes, the angle portion **535** may still extend into the gap **40** to fill the gap **40** and prevent the slurry from spreading outwards. Further, when the second adjustment portion **515** adopts a multi-stage elastic retractable structure, the second angle β may be adjusted by using the second elastic portion **5153**, thereby adjusting the first angle α, such that the first angle α matches the size of the gap **40**. Then, the first angle α is held to be matched with the gap **40** by the second limit portion **5155**. In this way, a relatively simple structure can be used to adjust the first angle α. When the second adjustment portion **515** adopts a multi-stage rigid retractable structure, the second angle β may be adjusted by using the extension and the retraction of the second mover **5159** relative to the second stator **5157**, thereby adjusting the first angle α, such that the first angle α matches the size of the gap **40** and hold the first angle α to match the gap **40**. In this way, a relatively simple structure can be used to adjust the first angle α.

[0076] Referring to FIGS. **2** and **4**, or FIGS. **3** and **4**, further, in some embodiments, the blocker **53** may also include a connection portion **539**. The connection portion **539** extends from the second surface **5317** of the cooperation portion **531**, and the connection portion **539** and the engagement portion **533** are located at a side of the cooperation portion **531** where the second surface **5317** of the cooperation portion **531** is located. The connection portion **539** is configured to connect to an external structure. The cooperation portion **531**, the engagement portion **533**, and the connection portion **539** together form an engagement groove **530**, and the support **51** is arranged in the engagement groove **530**. The connection portion **539** may have a block-like structure, an elongated structure, a rod-like structure, or a plate-like structure. The connection portion **539** is connected to the cooperation portion **531** at an end of the connection portion **539**, and another end of the connection portion **539** is a free end. The arrangement of the connection portion **539** can facilitate the fixed mounting of the entire blocker **53**. In addition, the engagement groove **530** formed by the cooperation portion **531**, the engagement portion **533**, and the connection portion **539** can provide a satisfactory mounting of the support **51**.

[0077] Referring to FIGS. **2** and **4**, or FIGS. **3** and **4**, further, in some embodiments, the coating device **100** may further include a loader **55** connected to the connection portion **539**.

[0078] In one embodiment, the loader **55** is fixed to a support (not shown) and is located above the slurry tank of the coating device **100**, and the connection portion **539** is mounted at the loader **55**. That is, the arrangement of the loader **55** is convenient for the mounting of the blocking assembly

50. As a result, the mounting of the blocking assembly **50** is simple and convenient. At the same time, the loader **55** can also prevent the slurry from overflowing. In addition, the loader **55** may be a block-like structure, an elongated structure, a rod-like structure, or a plate-like structure. Preferably, the loader **55** is structurally adapted to the connection portion **539**. For example, when the connection portion **539** has a block-like structure, the loader **55** also has a block structure; when the connection portion **539** has an elongated structure, the loader **55** also has an elongated structure; and when the connection portion **539** has a plate-like structure, the loader **55** also has a plate-like structure.

[0079] Referring to FIGS. **2** and **4**, or FIGS. **3** and **4**, further, in some embodiments, the coating device **100** may further include a clamping plate **57** configured to fixedly clamp the support **51** and the connection portion **539** to the loader **55**.

[0080] In one embodiment, the clamping plate **57** has a roughly V-shaped structure, and the clamping plate **57** includes a first clamping arm **571**, a second clamping arm **573**, and a connection arm **575** for connecting the first clamping arm **571** and the second clamping arm **573**. The first clamping arm **571** is clamped to the first support portion **511** of the support **51**, and the second clamping arm **573** is clamped to the loader **55**. The arrangement of the clamping plate **57** can firmly clamp the support **51** and the loader **55** without a movement. Therefore, the angle portion **535** of the blocker **53** can cooperate well with the coating roller **10** and the scraping roller **30**. Thus, the effect of blocking the overflowing of the slurry is improved.

[0081] Referring to FIG. **1**, the coating device **100** may further include a first support frame **61**, a first connector **71**, a second support frame **63**, and a second connector **73**. The scraping roller **30** is rotatably connected to the first support frame **61** through the first connector **71**. The coating roller **10** is rotatably connected to the second support frame **63** through the second connector **73**. In some embodiments, a positional relationship between the first support frame **61** and the second support frame **63** is an up-and-down arrangement, and a positional relationship between the first connector **71** and the second connector **73** is also an up-and-down arrangement. Each of the first support frame **61** and the second support frame **63** has a frame structure, and each of the first connector **71** and the second connector **73** is a connection shaft. Two ends of the scraping roller **30** are rotatably connected to two opposite ends of the first support frame **61** through the connection shafts, respectively, and two ends of the coating roller **10** are rotatably connected to two opposite ends of the second support frame **63** through the connection shafts, respectively. In this way, the scraping roller **30** is rotatable around the connecting shaft relative to the first support frame **61**, and the coating roller **10** is also rotatable around the connecting shaft relative to the second support frame **63**. In the embodiments of the present disclosure, the first support frame **61** and the first connector **71** jointly realize the mounting and the fixing of the scraping roller **30**, and the second support frame **63** and the second connector **73** jointly realize the mounting and the fixing of the coating roller **10**.

[0082] At last, it should be noted that, the above embodiments are only used to illustrate, rather than to limit, the technical solutions of the present disclosure. Although the present disclosure has been described in detail with reference to the above embodiments, those skilled in the art should understand that modifications may be made to the technical solutions described in the above embodiments, or equivalent replacements may be made to some or all of the technical features of the technical solutions described in the above embodiments. However, these modifications or replacements do not cause the essence of corresponding technical solutions to deviate from the scope of the technical solutions of the embodiments of the present disclosure, and shall be included in the scope of the claims and the specification of the present disclosure. In particular, the technical features mentioned in the various embodiments can be combined in any way as long as there are no structural conflicts. The present disclosure is not limited to the specific embodiments disclosed herein, but includes all technical solutions falling within the scope implementations the claims.

Claims

1. A coating device, comprising: a coating roller; a scraping roller, the scraping roller and the coating roller being arranged side by side, and a gap being provided between the scraping roller and the coating roller to allow a slurry to pass through; and a blocking assembly arranged at an outer side of the coating roller and an outer side of the scraping roller, wherein the blocking assembly comprises a support and a blocker mounted at the support, the blocker comprising a cooperation portion cooperating with the coating roller and an engagement portion cooperating with the scraping roller, wherein: a first end of the cooperation portion and a first end of the engagement portion are connected with each other and form an angle portion, the angle portion extending into the gap; and a second end of the cooperation portion is located away from a second end of the engagement portion to form a first angle between the cooperation portion and the engagement portion, the first angle being adjustable.
2. The coating device according to claim 1, wherein the blocker further comprises a first adjustment portion, two opposite ends of the first adjustment portion being connected to the cooperation portion and the engagement portion, respectively, and the first adjustment portion being configured to, by means of multi-stage extension and retraction, adjust the first angle according to a size of the gap.
3. The coating device according to claim 2, wherein the first adjustment portion comprises: a housing; an elastic portion penetrating the housing; and at least two limit portions mounted at the housing in an extendable and retractable manner, the at least two limit portions being sequentially arranged in a length direction of the elastic portion, and the at least two limit portions being configured to selectively extend to a deformation path of the elastic portion relative to the housing or retract to be located outside the deformation path of the elastic portion relative to the housing, to enable the first angle to match the gap.
4. The coating device according to claim 3, wherein the first adjustment portion adopts a multi-stage elastic retractable structure in such a manner that the first angle is adjusted through using the elastic portion.
5. The coating device according to claim 2, wherein the first adjustment portion comprises a stator and a mover, the mover being extendable and retractable relative to the stator to adjust the first angle to match the gap.
6. The coating device according to claim 5, wherein the first adjustment portion adopts a multi-stage rigid retractable structure in such a manner that the second angle is adjusted by using the extension and the retraction of the mover relative to the stator.
7. The coating device according to claim 1, wherein the support comprises: a first support portion connected to the engagement portion at a first end of the first support portion; and a second support portion connected to the cooperation portion at a first end of the second support portion, a second end of the first support portion and a second end of the second support portion being connected with each other to form a second angle, the second angle being adjustable to change the first angle between the cooperation portion and the engagement portion.
8. The coating device according to claim 7, wherein the support further comprises a second adjustment portion, two opposite ends of the second adjustment portion being connected to the first support portion and the second support portion, respectively, and the second adjustment portion being configured to, by means of multi-stage extension and retraction, adjust the second angle according to a size of the gap to change the first angle between the cooperation portion and the engagement portion.
9. The coating device according to claim 8, wherein the second adjustment portion comprises: a housing; an elastic portion penetrating the housing; and at least two limit portions mounted at the housing in an extendable and retractable manner, the at least two limit portions being sequentially

arranged in a length direction of the elastic portion, and the at least two limit portions being configured to selectively extend to a deformation path of the elastic portion relative to the housing or retract to be located outside the deformation path of the elastic portion relative to the housing, to enable the first angle to match the gap.

10. The coating device according to claim 9, wherein the second adjustment portion adopts a multi-stage elastic retractable structure in such a manner that the first angle is adjusted through using the elastic portion.

11. The coating device according to claim 8, wherein the second adjustment portion comprises a stator and a mover, the mover being extendable and retractable relative to the stator to adjust the first angle to match the gap.

12. The coating device according to claim 11, wherein the second adjustment portion adopts the multi-stage rigid retractable structure in such a manner that the second angle is adjusted by using the extension and the retraction of the mover relative to the stator.

13. The coating device according to claim 1, wherein: the cooperation portion has a first surface and a second surface opposite to the first surface of the cooperation portion, the first surface of the cooperation portion being an arc surface in contact with and in cooperation with the coating roller; and the engagement portion has a first surface and a second surface opposite to the first surface of the engagement portion, the first surface of the engagement portion being an arc surface in contact with and in cooperation with the scraping roller.

14. The coating device according to claim 13, wherein: the blocker further comprises a connection portion extending from the second surface of the cooperation portion; the connection portion and the engagement portion are located at a side of the cooperation portion where the second surface of the cooperation portion is located; the connection portion is configured to connect to an external structure; the cooperation portion, the engagement portion, and the connection portion together form an engagement groove; and the support is disposed in the engagement groove.

15. The coating device according to claim 14, wherein the connection portion has a block-like structure, an elongated structure, a rod-like structure, or a plate-like structure, the connection portion being connected to the cooperation portion at an end of the connection portion, and another end of the connection portion is a free end.

16. The coating device according to claim 14, further comprising a loader connected to the connection portion.

17. The coating device according to claim 16, wherein the loader is a block-like structure, an elongated structure, a rod-like structure, or a plate-like structure.

18. The coating device according to claim 16, further comprising a clamping plate configured to fixedly clamp the support and the connection portion to the loader.

19. The coating device according to claim 1, wherein the blocker is an ethylene-vinyl acetate copolymer, EVA, foam board, an acrylic board, a plastic board, or a metallic board.

20. The coating device according to claim 1, further comprising: a first support frame and a first connector, the scraping roller being rotatably connected to the first support frame through the first connector; and a second support frame and a second connector, the coating roller being rotatably connected to the second support frame through the second connector.